Comparative Performance of Lunar Earth-Departure Systems

Vehicle Mass, Propulsion Capability, and Mission Architecture

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September 28, 2026

Abstract

This working study compares past, current, and proposed lunar transportation architectures: Apollo's S-IVB/CSM/LM stack, SLS/ICPS with Orion and the European Service Module (ESM), Starship HLS, and Blue Moon Mark II. The analysis distinguishes mass delivered to low Earth orbit (LEO) from assembled mass at translunar injection (TLI). Where authoritative current engineering masses are unavailable, secondary-source values are explicitly treated as working estimates.

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1 Purpose and Method

At least two mass states must be distinguished: total mass delivered to a specified low Earth parking orbit and assembled vehicle mass at the beginning of TLI. Apollo provides a useful historical reference because the S-IVB, CSM, and LM departed Earth as an integrated lunar stack. Modern Artemis architectures distribute major mission elements across separate launch and propulsion chains.

The differences therein having completely separate vehicles in the Artemis architecture is a substantial departure from the simplicity of the Apollo stack. The mechanics of the management of the contract lunar vehicles, due to oversized vehicles, requires cryogenic refueling in orbit both for Blue Moon Mark II (cryogenic hydrogen and oxygen) and with Starship (cryogenic methane and oxygen: methalox for short).

Next we consider the flight paths and orbital mechanics of the differences between Apollo and Artemis. For example the rendezvous and docking of the lunar vehicle under the Artemis architecture is to happen in near-rectilinear halo orbit (NRHO) with a 6.5 day period around the moon; a much larger orbital path than Apollo's equatorial lunar orbit, leaving more the potential for variability and difficulty in orbital rendezvous and docking.

We consider the vehicle dynamics upon landing given vehicle center of mass. The Artemis architecture provides a much thinner margin for landing stability given the unstable equilibriums of the Blue Moon Mark II and the Starship; whereas, the Apollo LM has a much lower center of mass relative to its height, making landing in uneven terrain much more feasible.

Lastly, we consider the failure scenarios and resiliency of spacecraft: travel safety, propellant safety, accident resilience, and backup plans for rendezvous and docking on lunar return to command module.

2 Mass Accounting and Rocket Equation

Define m_D as dry stage/structural mass, m_U as propellant consumed, m_R as retained propellant, and m_L as delivered spacecraft/cargo. Then

$$m_0 = m_D + m_R + m_U + m_L, \qquad m_f = m_D + m_R + m_L.$$

The ideal rocket equation is

$$\Delta v = I_{\mathrm{sp}} g_0 \ln\left(\frac{m_0}{m_f}\right), \qquad g_0 = 9.806\,65\,\mathrm{m/s^2}.$$

Solving for useful delivered mass gives

$$m_L = \frac{m_U}{\exp\left(\Delta v/(I_{\mathrm{sp}} g_0)\right) - 1} - m_D - m_R.$$

Launch mass, parking-orbit mass, TLI ignition mass, and post-TLI mass are not interchangeable.

3 Numerical Vehicle Data

3.1 Apollo: S-IVB, CSM, and LM

Apollo 11 spacecraft masses used in this working comparison are:

Component	Mass (lb)	Mass (kg)
Command Module	12,250	5,557
Service Module	51,160	23,206
Lunar Module	33,200	15,059
Spacecraft–LM Adapter	4,045	1,835

For the LM, ascent-stage dry mass was about 4,804 lb, descent-stage dry mass 4,483 lb, RCS propellant 604 lb, descent propulsion propellant 18,100 lb, and ascent propulsion propellant 5,214 lb. Thus

$$m_{\mathrm{LM,dry}} \approx 4.21 \text{ t}, \qquad m_{\mathrm{LM,prop}} \approx 10.85 \text{ t}.$$

A working Saturn V 500-series S-IVB specification is:

Parameter	Working value
Empty mass	13.5 t
Propellant	109.5 t
Gross mass	123 t
Engine	1 J-2
Vacuum thrust	1,033.1 kN
Vacuum specific impulse	421 s
Propellants	LOX/LH_2

For Apollo 11, the as-flown mission ledger gives the complete vehicle at TLI ignition as

$$M_{\mathrm{Apollo,TLI,start}} = 296,379 \text{ lb} \approx 134.44 \text{ t}.$$

Approximately 164,431 lb (74.58 t) of S-IVB propellant was consumed during the principal TLI burn interval. The complete remaining vehicle mass at TLI completion was approximately

$$M_{\mathrm{Apollo,TLI,end}} = 139,369 \text{ lb} \approx 63.22 \text{ t}.$$

3.2 SLS, ICPS, Orion, and ESM

Working Artemis I values are:

Parameter	Working value
Orion gross launch mass	32.65 t
Orion mass at TLI	24.05 t
Post-TLI Orion mass	23.85 t
Orion usable propellant	7.25 t
ESM dry mass (secondary-source working value)	6.185 t
ESM fueled/launch mass (configuration dependent)	15.461 t
ESM usable propellant (working value)	8.6 t

The ICPS is represented approximately by:

ICPS parameter	Working value
Dry mass	~ 3.8 t
Fully loaded mass	~ 32–33 t
Propellant	~ 28–29 t
Engine	1 RL10B-2
Vacuum thrust	~ 110 kN
Propellants	LOX/LH_2

A rough launch-configuration upper-stack mass is therefore of order

$$M_{\text{ICPS+adapter+Orion}} \sim 66 \text{ t},$$

depending on exact mission configuration. This is not the TLI ignition mass: ICPS propellant is consumed before TLI and Orion sheds ascent hardware. A burn-by-burn ledger is required.

3.3 Starship Human Landing System

Where a current authoritative HLS-specific dry mass is unavailable, this working study uses the commonly cited reference-Starship values as a proxy:

Parameter	Working value
Dry mass proxy	~ 100 t
Propellant capacity	$\sim 1,200$ t
Fully fueled mass proxy	$\sim 1,300$ t
Propellants	LOX/CH_4
HLS payload capability	100+ t class

The 100-t dry value is a secondary-source/reference-Starship proxy, not a measured HLS flight dry mass. More recent generic Starship specifications have quoted approximately 1,600 t of spacecraft propellant; that represents a different revision and should be treated as a separate sensitivity case.

For scale,

$$\frac{1,300}{134.44} \approx 9.67.$$

Thus the fully fueled 1,300-t proxy is about 9.7 times the mass of the entire Apollo 11 vehicle at TLI ignition. This does not establish efficiency; it shows why all tanker/refueling mass needed to establish that departure state must be counted.

3.4 Blue Moon Mark II

Secondary-source values are used where a complete current engineering mass breakdown is not public:

Parameter	Working value
Dry mass	~ 16 t
Launch/wet mass	> 45 t
Loaded-minus-dry mass	> 29 t
Height	~ 16 m
Main engines	3 BE-7
Propellants	LOX/LH_2

The > 29 t difference must not automatically be identified entirely as usable propellant because consumables and configuration-dependent hardware may be included. The associated Blue Origin Transporter architecture has been described as capable of moving approximately 100 t of propellant from Earth orbit toward lunar orbit.

4 Summary Mass Table

System	Dry/hardware	Propellant	Loaded/relevant mass	Status
Apollo S-IVB	13.5 t	109.5 t	123 t	Flown/reference
Apollo 11 LM	4.21 t	10.85 t	15.06 t	Flown
Apollo 11 complete TLI stack	—	—	134.44 t	Flown
ICPS	~ 3.8 t	~ 28–29 t	~ 32–33 t	Flown
Orion spacecraft	—	7.25 t usable	24.05 t at TLI	Flown
ESM	6.185 t	8.6 t working	15.461 t	Mixed/config.-dependent
Starship/HLS proxy	~ 100 t	$\sim 1,200$ t	$\sim 1,300$ t	Dev/proxy
Blue Moon Mk2	~ 16 t	> 29 t loaded-minus-dry	> 45 t	Dev/secondary

5 Architecture-Level Transport Efficiency

For architectures requiring multiple launches,

$$M_{\mathrm{LEO,total}} = \sum_{i=1}^{N} M_{\mathrm{LEO},i}.$$

Define the study-specific metric

$$\eta_{\mathrm{TLI}} = \frac{m_{\mathrm{useful,TLI}}}{M_{\mathrm{LEO,total}}}.$$

For orbital-refueling architectures, the denominator includes supporting propellant/tanker launches. This is not a standard universal rocket-performance parameter; it is defined here to expose architecture-level mass efficiency.

6 Normalized Versus Mission-Native Analysis

A normalized comparison places each departure system conceptually in the same reference parking orbit and assigns a common target departure condition. This isolates dry mass, propellant, specific impulse, thrust, payload, and burn duration.

A mission-native comparison instead preserves actual parking orbit, ascent expenditure, orbital assembly, refueling, spacecraft configuration, reserves, and departure sequence. The first asks how capable the propulsion stage is; the second asks how capable the complete transportation architecture is.

7 Apollo and Artemis as Different Mass Architectures

Apollo 11 began TLI with approximately 134.44 t of integrated vehicle mass, including the Earth-departure stage, CSM, LM, and structural hardware. Artemis I Orion itself was approximately 24.05 t at TLI. These are not equivalent system quantities because Orion's TLI mass does not contain the lunar lander.

For a landing architecture, a useful schematic accounting is

$$M_{\text{Artemis,total}} = M_{\text{Orion chain}} + M_{\text{lander chain}} + M_{\text{refueling/transport chain}},$$

with careful treatment of docking and propellant transfer to prevent double counting.

7.1 Vehicle Dynamics Beyond Mass Ratio

Mass ratio is a fundamental parameter in launch-vehicle analysis, but by itself it does not completely characterize the capability of a vehicle or upper stage. Two stages may have identical mass ratios while providing substantially different mission performance because of differences in specific impulse, thrust, burn duration, structural mass, staging, and trajectory. For lunar missions in particular, the relevant question is not simply how much propellant a vehicle carries, but how much useful mass the vehicle can move from one required orbital state to another.

A useful vehicle-dynamics analysis therefore begins with the rocket equation and extends it to include propulsion performance, finite-burn effects, orbital mechanics, staging, and mission-specific Δv requirements.

7.1.1 Mass Ratio and Propellant Fraction

For a single stage, the mass ratio is

$$MR = \frac{m_0}{m_f}, \tag{1}$$

where m_0 is the mass at ignition and m_f is the mass following the burn. The initial mass can be decomposed as

$$m_0 = m_{\text{dry}} + m_p + m_L, \tag{2}$$

where m_{dry} is the stage dry mass, m_p is the usable propellant mass, and m_L represents payload and any downstream hardware that must also be accelerated.

Following propellant depletion,

$$m_f = m_{\text{dry}} + m_L. \tag{3}$$

It is also useful to define the propellant mass fraction,

$$\lambda_p = \frac{m_p}{m_0}, \tag{4}$$

and the structural coefficient,

$$\epsilon_s = \frac{m_{\text{dry}}}{m_{\text{dry}} + m_p}. \tag{5}$$

The structural coefficient permits stages of substantially different physical size to be compared on the basis of the fraction of stage mass devoted to hardware rather than propellant.

7.1.2 Specific Impulse and Exhaust Velocity

Mass ratio becomes dynamically meaningful only when combined with propulsion performance. The effective exhaust velocity is

$$v_e = I_{sp} g_0, \tag{6}$$

where I_{sp} is the engine specific impulse and

$$g_0 \approx 9.80665 \text{ m/s}^2. \tag{7}$$

Consequently, two stages having the same mass ratio can have substantially different velocity capability if their propulsion systems have different specific impulses. This distinction is especially important when comparing high-specific-impulse cryogenic upper stages with lower-specific-impulse propulsion systems.

7.1.3 Ideal Delta-V

The ideal velocity increment available from a stage is given by the Tsiolkovsky rocket equation,

$$\Delta v_{\text{ideal}} = I_{sp} g_0 \ln\left(\frac{m_0}{m_f}\right). \tag{8}$$

Substituting the individual mass components gives

$$\Delta v_{\text{ideal}} = I_{sp} g_0 \ln\left(\frac{m_{\text{dry}} + m_p + m_L}{m_{\text{dry}} + m_L}\right). \tag{9}$$

For a multistage vehicle, the total ideal velocity increment is obtained by summing the contribution of each stage,

$$\Delta v_{\text{ideal,total}} = \sum_i I_{sp,i} g_0 \ln\left(\frac{m_{0,i}}{m_{f,i}}\right). \tag{10}$$

Care must be taken when applying this expression because the payload of one stage may include the complete mass of all subsequent stages and spacecraft.

7.1.4 Payload Sensitivity

The rocket equation can also be inverted to determine the payload that a stage can accelerate through a specified velocity increment. Let

$$R = \exp\left(\frac{\Delta v}{I_{sp} g_0}\right). \tag{11}$$

Then

$$R = \frac{m_{\text{dry}} + m_p + m_L}{m_{\text{dry}} + m_L}. \tag{12}$$

Solving for payload gives

$$m_L = \frac{m_p}{R - 1} - m_{\text{dry}}, \tag{13}$$

or equivalently,

$$\boxed{m_L = \frac{m_p}{\exp\left(\Delta v/(I_{sp} g_0)\right) - 1} - m_{\text{dry}}}. \tag{14}$$

This relationship is particularly useful for comparing lunar architectures. Rather than asking only how massive an upper stage is, one can ask how much useful spacecraft mass the stage can inject onto a specified trajectory.

7.1.5 Thrust-to-Weight Ratio

The rocket equation does not contain thrust explicitly. Nevertheless, thrust has an important effect on actual vehicle performance. The instantaneous thrust-to-weight ratio is

$$\text{TWR} = \frac{T}{mg}, \tag{15}$$

where T is total engine thrust and m is instantaneous vehicle mass.

During ascent from the Earth's surface, thrust-to-weight ratio strongly affects gravity losses. For an upper stage already in orbit, the effect is different because the vehicle is already in free fall. Nevertheless, thrust determines burn duration, and a sufficiently long burn can no longer be treated as an instantaneous or impulsive orbital maneuver.

7.1.6 Propellant Flow Rate and Burn Duration

For approximately constant thrust and specific impulse, propellant mass flow is

$$\dot{m} = \frac{T}{I_{sp} g_0}. \tag{16}$$

An approximate burn duration is therefore

$$t_b = \frac{m_p}{\dot{m}}. \tag{17}$$

Burn duration is particularly important for translunar injection. An ideal orbital maneuver is often modeled as an instantaneous change in velocity. A real TLI maneuver instead occurs over a finite arc of the parking orbit.

A high-thrust upper stage can concentrate a larger fraction of its energy addition near the desired point in the orbit. A lower-thrust stage may require a longer burn, requiring numerical trajectory integration rather than a purely impulsive approximation.

7.1.7 Ideal Delta-V Versus Effective Delta-V

The rocket equation describes the ideal velocity capability of the propulsion system. Actual trajectory performance can be represented schematically as

$$\Delta v_{\text{effective}} = \Delta v_{\text{ideal}} - \Delta v_{\text{gravity}} - \Delta v_{\text{drag}} - \Delta v_{\text{steering}} - \Delta v_{\text{other}}. \tag{18}$$

For launch from the Earth's surface, atmospheric drag and gravity losses can be significant. For an upper stage performing TLI from an established parking orbit, atmospheric drag is effectively absent, but finite-burn, steering, guidance, and other trajectory effects remain.

Thus, quoted ideal Δv should not automatically be interpreted as mission-usable Δv.

7.1.8 Orbital Energy

Velocity alone is not sufficient to characterize a lunar trajectory because the significance of a particular velocity depends upon position within the gravitational field. A more useful quantity is specific orbital mechanical energy,

$$\mathcal{E} = \frac{v^2}{2} - \frac{\mu}{r}, \tag{19}$$

where μ is the gravitational parameter of the central body and r is distance from its center. For a Keplerian orbit,

$$\mathcal{E} = -\frac{\mu}{2a}, \tag{20}$$

where a is the semimajor axis. Combining these relations produces the vis-viva equation,

$$v^2 = \mu\left(\frac{2}{r} - \frac{1}{a}\right). \tag{21}$$

This provides the connection between propulsion performance and trajectory. A propulsion system supplies Δv, while the resulting change in velocity changes the spacecraft's orbital energy and therefore its orbit.

7.1.9 Translunar Injection

For a simplified patched-conic treatment, a spacecraft initially in a circular Earth parking orbit has velocity

$$v_{\mathrm{LEO}} = \sqrt{\frac{\mu_E}{r_{\mathrm{LEO}}}}. \tag{22}$$

If the translunar trajectory is approximated as an elliptical transfer orbit having perigee at the parking orbit and apogee near lunar distance, then

$$a_{\mathrm{TLI}} = \frac{r_{\mathrm{LEO}} + r_{\mathrm{Moon}}}{2}. \tag{23}$$

The required velocity at transfer-orbit perigee is

$$v_{\mathrm{TLI}} = \sqrt{\mu_E\left(\frac{2}{r_{\mathrm{LEO}}} - \frac{1}{a_{\mathrm{TLI}}}\right)}. \tag{24}$$

The corresponding ideal impulsive TLI requirement is therefore

$$\Delta v_{\mathrm{TLI}} = v_{\mathrm{TLI}} - v_{\mathrm{LEO}}. \tag{25}$$

This quantity can be compared directly with the available Δv of an upper stage carrying a specified spacecraft stack.

A more accurate lunar trajectory calculation must include the Moon's motion, three-body effects, departure geometry, launch date, inclination, targeting requirements, and finite burn duration. Nevertheless, the patched-conic calculation provides a useful first-order comparison between candidate vehicles.

7.1.10 Delta-V Margin

For each major mission maneuver, the relevant engineering quantity is the difference between available and required velocity capability,

$$\Delta v_{\text{margin}} = \Delta v_{\text{available}} - \Delta v_{\text{required}}. \tag{26}$$

A positive ideal margin is necessary but is not, by itself, sufficient to establish mission feasibility. The margin must also accommodate trajectory corrections, guidance errors, dispersions, propellant reserves, residual propellant, and other operational contingencies.

Consequently, a configuration that satisfies the nominal maneuver by only a very small theoretical margin should not be considered equivalent to one having substantial operational reserve.

7.1.11 Staging and Discarded Mass

A complete vehicle analysis must also account for when hardware is discarded. The importance of a kilogram of mass depends strongly upon how far through the mission it must be transported.

For example, a kilogram discarded before TLI does not need to be accelerated through TLI. A kilogram retained through TLI but discarded before lunar descent does not need to be landed. Conversely, a kilogram that travels to the lunar surface and subsequently returns must potentially be accelerated through several major maneuvers.

For this reason, the mass model should be evaluated immediately before and after every major staging event:

$$m_i^- \longrightarrow m_i^+, \tag{27}$$

where m_i^- denotes vehicle mass immediately before separation and m_i^+ denotes the remaining vehicle mass immediately afterward.

This staging analysis is especially important when comparing architectures whose total launch masses are very different but whose masses delivered to the lunar vicinity or lunar surface may be much closer.

7.1.12 Usable Propellant, Reserves, and Residuals

Nominal tank capacity should not automatically be treated as available propellant. It is useful to distinguish

$$m_{p,\text{loaded}} = m_{p,\text{usable}} + m_{p,\text{reserve}} + m_{p,\text{residual}}. \tag{28}$$

Propellant may be unavailable for nominal mission maneuvers because of engine shutdown requirements, trapped residuals, ullage requirements, guidance reserves, boiloff, thermal conditioning, or mission contingency requirements.

For cryogenic stages, elapsed time can additionally make propellant mass a time-dependent quantity,

$$m_p = m_p(t), \tag{29}$$

which becomes increasingly important for architectures requiring long loiter periods, propellant transfer, or multiple rendezvous operations.

7.1.13 Mission-State Mass and Energy Ledger

A useful method for comparing lunar architectures is therefore to construct a mass and energy ledger at each major mission state. A generic ledger has the form

Mission state	Mass	Propellant	Velocity	Specific energy	Required Δv	Remaining Δv
LEO insertion	m_1	$m_{p,1}$	v_1	$\mathcal{E}_1$	—	Δv_1
Before TLI	m_2	$m_{p,2}$	v_2	$\mathcal{E}_2$	Δv_{TLI}	Δv_2
After TLI	m_3	$m_{p,3}$	v_3	$\mathcal{E}_3$	—	Δv_3
Lunar arrival	m_4	$m_{p,4}$	v_4	$\mathcal{E}_4$	Capture/LOI	Δv_4
Before descent	m_5	$m_{p,5}$	v_5	$\mathcal{E}_5$	Descent	Δv_5
Lunar surface	m_6	$m_{p,6}$	—	—	Ascent	Δv_6
Before Earth return	m_7	$m_{p,7}$	v_7	$\mathcal{E}_7$	TEI	Δv_7

Such a ledger makes explicit what mass is present, what propellant remains, what orbital state has been achieved, and what propulsion capability remains at every important point in the mission.

7.1.14 Implications for Vehicle Comparison

The resulting comparison is therefore not simply one of gross vehicle mass or propellant capacity. A useful lunar-vehicle comparison should include at least

$$\{m_{\mathrm{dry}}, m_p, m_L, I_{sp}, T, t_b, \Delta v, \mathcal{E}, \epsilon_s\}\,, \tag{30}$$

together with the sequence of staging events and the Δv requirements of the selected trajectory.

This distinction is important when comparing vehicles such as the S-IVB, ICPS, EUS, Apollo spacecraft, and proposed lunar lander systems. A larger vehicle does not necessarily deliver proportionally more useful lunar payload, while a vehicle with a favorable mass ratio does not necessarily have sufficient propulsion or trajectory capability to perform the required mission.

The more physically meaningful question is therefore not simply

> How much does the vehicle weigh?

but rather

> Given its dry mass, propellant, propulsion system, staging sequence, and trajectory, how much useful spacecraft mass can the vehicle move from one required orbital state to the next while maintaining adequate mission margin?

This mass–energy–trajectory formulation provides a common basis on which historical Apollo hardware, current SLS-derived systems, and proposed lunar architectures can be compared.

8 Performance Versus Survivability

Reliability, redundancy, abort capability, and crew survivability are important architectural questions but are deliberately separated from the mass/performance analysis. This prevents propulsion efficiency from being conflated with failure tolerance. A later section can compare independent propulsion capability, docked lifeboat capability, consumables, abort trajectories, and recovery options.

9 Planned Orbital-Mechanics Comparison

The second major part of the study will compare Apollo and Artemis flight paths on a maneuver-by-maneuver basis: parking orbit, Earth departure, outbound corrections, lunar capture or flyby, staging orbit, rendezvous, descent, ascent, return departure, and Earth entry. Each event should record responsible propulsion system, pre- and post-burn mass, propellant expenditure, Δv, elapsed time, and remaining contingency capability.

10 Source Treatment

Flown mission values and primary NASA/contractor specifications should be preferred. When such a value is unavailable for a developing vehicle, this working document permits Wikipedia or another secondary compilation as an explicitly identified estimate. No secondary estimate should be silently represented as a measured or finalized engineering specification. Vehicle revision and source date should accompany configuration-sensitive values.